# Literal Resonance after TPC-101: Verified Branch Closures, the Three-Ledger Master Inequality, and a Stop–Go Roadmap

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**Abstract**

TPC-93 through TPC-101 develop a literal low-frequency route for one prescribed nonzero shift $h_0$. The sequence contains genuine branch closures, exact finite-group reductions, and several sharp warnings against false shortcuts. This paper audits the combined proof state and gives one consolidated inequality for the remaining positive resonant carrier.

After the separate $q_X \mid u$ return, let $M_H^{\mathrm{inv}} = \|a_H^{\mathrm{inv}}\|_1$ be the remaining invertible frequency-free multiplier census, let

$$g_q(H) = \sqrt{\frac{2H(q-1-2H)}{q-1}},$$

and define

$$\mathfrak{P}_X = \sum_H \frac{2H}{q_X - 1} M_H^{\mathrm{inv}}, \qquad \mathfrak{X}_X = \sum_H g_{q_X}(H) \sqrt{|\mathfrak{X}_H^\circ|}.$$

Here $\mathfrak{X}_H^\circ$ is the exact centered cross-cell excess from TPC-100. If $W_X$ is the maximum normalized opened-atom weight and $\mathcal{H}_X$ is the set of active nonprincipal widths, then TPC-98, TPC-100, and TPC-101 imply the master bound

$$\boxed{\mathcal{C}_{\mathrm{exc},X}^{\mathrm{abs}} \ll X^{o(1)} Q_X^2 + B_X \Big[\mathfrak{P}_X + \sqrt{W_X |\mathcal{H}_X| (q_X - 1) \mathfrak{P}_X} + \mathfrak{X}_X\Big]},$$

where $B_X = \|b_X\|_1 \ll_K 1$.

Thus the positive resonance path now has three explicit literal obligations: principal mass, the opened-atom cap, and width-weighted cross-cell dispersion. Conductor-one keys and the nonconstant $q_X \mid u$ branch are already returned at $X^{o(1)} Q_X^2$. This is real L1 progress, but not an L2 arithmetic theorem. Generic signed affine Möbius cancellation, TPC-18 full-block reassembly, determinant/zero-mode compatibility, and the strict physical endpoint budget remain independent gaps.

We give proof-carrying stop criteria and a next-batch order: first audit the atom cap, then principal mass, then canonical cross-map collisions. A proof-carrying polynomial obstruction in an actual term of the master inequality switches the program to the signed filter-before-majorant route; it is not relabeled as evidence for a prime-pair theorem.

**Keywords:** fixed-shift Möbius correlation; literal resonance; proof audit; multiplier census; cross-cell dispersion; parity barrier; stop–go roadmap.

**Audit convention.** Every statement is labeled by what it actually proves: L0 for finite algebra or model identities, L1 for literal physical-frame certificates and conditional reassembly, and L2 only for a growing fixed nonzero-$h_0$ arithmetic estimate. No support-only, averaged, frozen-family, or surrogate gain is reported as L2.

# 1 Scope and nonnegotiable invariants

The object under discussion is one growing physical coefficient with a prescribed integer

$$h_0 \neq 0.$$

The shift is never specialized to 2. A theorem uniform in such $h_0$ may later be evaluated at 2, but a surrogate average in $h$ is not fixed-$h_0$ progress.

## 1.1 Three proof levels

L0 Finite algebra, exact reindexing, group spectra, model inequalities, and counterexamples. These may identify a valid door but do not estimate the growing arithmetic coefficient.

L1 Proof-carrying return to the literal physical packet: exact source keys, actual support and masks, both polarizations, all exact-content and projector children, and one global normalization.

L2 A new estimate for the actual growing fixed-$h_0$ arithmetic quantity, strong enough to enter the final proof ledger.

## 1.2 Literal data that may not be quotiented away

Every positive or signed route must preserve:

1. the two physical rows and determinant orientation;
2. the opened divisor and complementary factor at the atom where they actually occur;
3. $c, \kappa, b = c\kappa, g = (b, \sigma)$, and $B = b/g$ as distinct exact-content data;
4. the least-absolute signed low-frequency lift and the true phase modulus;
5. every interval, residue, prefix, local-mask, row-gcd, and projector label;
6. the actual active support rather than a rectangular ambient box; and
7. the original global normalization, exactly once.

The corrected TPC-99 census therefore uses the frequency-free key $\beta = (\theta, c, \kappa)$ and only then pairs it with the actual frequency weight. A key already containing $r$ cannot be convolved with all $r$ again.

## 1.3 Independent exponent ledgers

The compatibility

$$\lambda_D \leq 2\eta_Z$$

belongs to determinant energy and zero-mode assembly. The condition

$$\Lambda_{\text{phys}} < \frac{1}{400}$$

belongs to the final physical loss. They are not interchangeable, and equality at the latter endpoint is a stop. A spectral saving already present inside an analytic theorem may not be charged a second time as a physical gain.

*Remark* 1.1 (Audit target). This paper audits the *resonant exceptional carrier*. Closing it would not close the generic affine sector or the full TPC-18 block. The scope restriction is part of the theorem, not a disclaimer added after the proof.

| Paper | Verified advance | Level | What remains |
|---|---|---|---|
| TPC-93 | Exact low-window export; explicit source–child inverse, projector multiplicity, both polarizations, and tail return | L0/L1 | No growing affine cancellation estimate |
| TPC-94 | Exact-content progression; determinant $Bh_0 \to h_0$; literal conductor 1 or $q_X$; exact fiber length | L0/L1 | Exceptional branch mass and nonconstant resonance |
| TPC-95 | Sharp free-phase dominance energy, target rigidity, and compressed collision census | L0/L1 | Actual determinant lower bound and loss $< 1$ |
| TPC-96 | Exact anchored Tauberian identity and translation-defect criterion | L0 | The required uniform prefix input; averages alone do not suffice |
| TPC-97 | Constant-preserving projection monotonicity and exact defect when constants are not preserved | L0/L1 | Projection cannot manufacture normalized zero-mode gain |
| TPC-98 | Full literal conductor-one branch returned at $X^{o(1)}Q_X^2$ | L1 | No obligation remains for that branch |
| TPC-99 | Exact multiplicative resonance spectrum; corrected frequency-free positive census | L0/L1 | Principal and centered census gates |
| TPC-100 | The $q_X \mid u$ nonconstant branch returned; invertible branch linearized; exact singleton-cell diagonal/cross-cell split | L0/L1 | Fine-cell injectivity is tautological; all distinct-atom collisions remain in the cross ledger |
| TPC-101 | Fixed-filter norm; $\sqrt{q}\log q$ replaced by exact $g_q(H)$; principal-to-diagonal transfer under atom cap | L0/L1 | Principal mass, literal atom cap, width-weighted cross excess |

Table 1: Proof-state audit. No row in the table is an L2 fixed-shift arithmetic theorem.

# 2 Dependency audit: TPC-93 through TPC-101

## 2.1 Two foundations repaired rather than assumed

The TPC-93 lossless export now contains an explicit source–child map and inverse, rather than citing an anonymous decorated ledger. Its singleton statement also distinguishes algebraic conductor one from a character that is merely constant on one observed point [2].

The TPC-99 census is indexed by $\beta = (\theta, c, \kappa)$, not by $(\theta, c, \kappa, r)$. Its weight

$$w_{\beta,X} = |A_{\beta,X}|\mathcal{A}_{\beta,X}$$

is uniform in $r$, and $N_\beta$ is the resolved $z$-fiber cardinality. This prevents a second frequency cross-product and keeps the global normalization literal [3, 7].

## 2.2 Useful negative information

TPC-96 rules out a common Tauberian shortcut: a small logarithmic primitive on a translated interval yields the desired terminal average only when the anchor defect has the required scale [5]. TPC-97 rules out another shortcut: constant-preserving orthogonal coarsening can only improve a zero-mode-to-energy ratio; a nonconstant-preserving projection needs its explicit defect budget [6]. These are not failed papers. They remove invalid proof paths before polynomial losses are hidden in them.

*Remark* 2.1 (The dominance branch remains available). TPC-95 proves an exact dominance-energy identity and a sharp shared-target rigidity bound [4]. It remains a parallel signed/energy route, but its conditional lower-energy input has not become an arithmetic theorem merely because the resonance route advanced.

## 3 The three-ledger master inequality

Let $\mathcal{C}^{\mathrm{abs}}_{\mathrm{exc},X}$ denote the positive literal resonant carrier after the TPC-93–TPC-94 low-window export. It is a majorant for the signed resonant contribution, not an identity with it.

TPC-98 and TPC-100 give the disjoint split

$$\mathcal{C}^{\mathrm{abs}}_{\mathrm{exc},X} = \mathcal{C}^{\mathrm{abs}}_{\mathrm{const},X} + \mathcal{C}^{\mathrm{abs}}_{q|u,X} + \mathcal{C}^{\mathrm{abs}}_{\mathrm{inv},X},$$

where

$$\mathcal{C}^{\mathrm{abs}}_{\mathrm{const},X} \ll X^{o(1)} Q_X^2, \tag{1}$$
$$\mathcal{C}^{\mathrm{abs}}_{q|u,X} \ll X^{o(1)} Q_X^2. \tag{2}$$

The remaining term has $q_X \nmid u$ and the exact cellwise multiplier

$$\omega = A(mj + h_0) \quad \text{in } \mathbb{F}_{q_X}^{\times}.$$

For each exact resonance width $H$ on the remaining invertible branch, write

$$M_H^{\mathrm{inv}} = \|a_H^{\mathrm{inv}}\|_1, \qquad \|(a_H^{\mathrm{inv}})^\circ\|_2^2 = \mathfrak{D}_H^\circ + \mathfrak{X}_H^\circ.$$

Define

$$B_X = \|b_X\|_1, \tag{3}$$
$$\mathfrak{P}_X = \sum_H \frac{2H}{q_X - 1} M_H^{\mathrm{inv}}, \tag{4}$$
$$\mathfrak{X}_X = \sum_H g_{q_X}(H)\sqrt{|\mathfrak{X}_H^\circ|}, \tag{5}$$
$$W_X = \max_{H,C,j} w_C(j), \tag{6}$$
$$\mathcal{H}_X = \{H : g_{q_X}(H) > 0,\ M_H^{\mathrm{inv}} > 0\}. \tag{7}$$

The maximum is taken only over the actual $q_X \nmid u$ opened-atom support after one global normalization.

**Theorem 3.1** (Three-ledger master inequality)**.** *Assume the literal source-to-branch and branch-to-opened-atom crosswalks of TPC-93–TPC-100. Then*

$$\mathcal{C}^{\mathrm{abs}}_{\mathrm{exc},X} \ll X^{o(1)} Q_X^2 + B_X \left[\mathfrak{P}_X + \sqrt{W_X |\mathcal{H}_X| (q_X - 1)\mathfrak{P}_X} + \mathfrak{X}_X\right]. \tag{8}$$

*Proof.* The first term contains (1)–(2). On the invertible branch, TPC-101 gives

$$\mathcal{C}^{\mathrm{abs}}_{\mathrm{inv},X} \le B_X \mathfrak{P}_X + B_X \sum_H g_{q_X}(H) \|(a_H^{\mathrm{inv}})^\circ\|_2.$$

The TPC-100 centered decomposition implies

$$\|(a_H^{\mathrm{inv}})^\circ\|_2 \le \sqrt{\mathfrak{D}_H^\circ} + \sqrt{|\mathfrak{X}_H^\circ|}.$$

The second sum is $\mathfrak{X}_X$. For the first, each diagonal ledger obeys

$$0 \le \mathfrak{D}_H^\circ \le \sum_{C,j} w_C(j)^2 \le W_X M_H^{\mathrm{inv}}.$$

Cauchy–Schwarz over $\mathcal{H}_X$, together with

$$g_{q_X}(H)^2 \le 2H,$$

gives

$$\sum_H g_{q_X}(H)\sqrt{\mathfrak{D}_H^\circ} \le \sqrt{W_X |\mathcal{H}_X| (q_X - 1)\mathfrak{P}_X}.$$

Combining the three displays proves (8). □

**Corollary 3.2** (Conditional closure of positive resonance)**.** *Suppose*

$$q_X \asymp Q_X, \qquad B_X \ll_K 1,$$

*and*

$$\mathfrak{P}_X \le X^{o(1)} Q_X^2, \tag{9}$$

$$W_X \le X^{o(1)}, \tag{10}$$

$$\mathfrak{X}_X \le X^{o(1)} Q_X^2. \tag{11}$$

*Then*

$$\mathcal{C}^{\mathrm{abs}}_{\mathrm{exc},X} \ll X^{o(1)} Q_X^2.$$

*Proof.* Since $|\mathcal{H}_X| \le (q_X - 1)/2$, the middle radical in (8) is

$$\le X^{o(1)} \sqrt{q_X^2 Q_X^2} \ll X^{o(1)} Q_X^2.$$

The remaining terms have the same bound. □

*Remark* 3.3 (Sufficient, not necessary). Corollary 3.2 closes a positive-majorant route. Failure of one obligation does not imply a large signed resonant sum; cancellation discarded in forming $\mathcal{C}^{\mathrm{abs}}_{\mathrm{exc},X}$ may still be decisive.

# 4 What the positive route must now prove

## 4.1 Gate P: principal mass

The exact obligation is

$$\mathfrak{P}_X = \sum_H \frac{2H}{q_X - 1} M_H^{\mathrm{inv}} = \sum_{\substack{\beta \in \mathfrak{B}_X^{\mathrm{nc}} \\ 1 \le N_\beta \le q_X}} \frac{2H_{q_X}(N_\beta)}{q_X - 1} \sum_{\substack{z \in I_\beta^* \\ q_X \nmid u_{\beta,z}}} w_{\beta,z} \le X^{o(1)} Q_X^2.$$

For a branch of resolved $z$-fiber length $N_\beta \ge 3$,

$$\frac{H_{q_X}(N_\beta)}{q_X} \le \frac{1}{N_\beta}.$$

This suggests opening the positive mass before grouping by $H$ and proving an inverse-length estimate. It does *not* justify replacing $M_H^{\mathrm{inv}}$ by an ambient count: $M_H^{\mathrm{inv}}$ contains the literal row coefficients, content and projector masses, masks, and one global normalization.

After the $q_X \mid u_{\beta,z}$ atoms have been returned, the remaining invertible endpoint atoms with $N_\beta = 1, 2$ are purely principal. They receive no centered TPC-101 cost, so they must be returned entirely through $\mathfrak{P}_X$. Calling them “short” does not make them negligible.

## 4.2 Gate W: maximum opened-atom weight

TPC-101 needs

$$W_X = \max w_C(j) \le X^{o(1)}$$

to derive the diagonal gate from Gate P. This must be checked by writing one atom literally:

$$w_C(j) = |\mathsf{R}_C(j)\, \mathsf{C}_C(j)\, \mathsf{S}_C(j)\, \mathsf{N}_X|\,. \tag{12}$$

Here $\mathsf{R}_C$ contains the row coefficients, $\mathsf{C}_C$ the content and projector coefficient, $\mathsf{S}_C$ the local and smooth weights, and $\mathsf{N}_X$ the single global normalization. An $\ell^1$ bound for all rows does not by itself imply the desired pointwise cap after a different normalization. Conversely, if each factor in (12) has a subpower pointwise envelope, Gate W is a local bookkeeping theorem rather than a new Möbius-cancellation theorem.

### 4.3 Gate X: genuine cross-cell excess

TPC-100 gives the exact collision support

$$A_C m_C j - A_D m_D k \equiv h_0(A_D - A_C) \pmod{q_X}.$$

The remaining ledger is

$$\mathfrak{X}_X = \sum_H g_{q_X}(H)\sqrt{|\mathfrak{X}_H^\circ|}.$$

At the fully literal TPC-100 resolution, retaining $(m, u, \sigma)$ already recovers $j = (\sigma u - h_0)/m$, so every fine cell is a singleton. The resulting same-cell "injectivity" is therefore tautological and supplies no dispersion gain. All collisions between distinct opened atoms—including atoms from the same parent $\beta$—stay in the cross-cell term, and different cells may also induce the same affine map. A valid proof should therefore:

1. aggregate identical maps only after retaining a recoverable provenance bag and exact total weight;
2. separate identical-map mass from genuinely distinct-map collisions;
3. center with the physical baseline $M_C M_D/(q_X - 1)$; and
4. use the width factor $g_{q_X}(H)$ before summing over $H$.

**Proposition 4.1** (Correct order of attack)**.** *Within the positive route, Gate W should be audited before a new diagonal-energy theorem, and Gate P before a global cross-cell large sieve.*

*Proof.* If Gates P and W hold, the diagonal term already follows from theorem 3.1; a separate diagonal theorem would duplicate work. If the actual principal term $B_X \mathfrak{P}_X$ fails polynomially, centered dispersion cannot repair the *termwise upper bound* in theorem 3.1, so a cross-cell theorem alone would not close this three-ledger route. A sharper filter-sensitive positive theorem could still use cancellation between the principal and nonprincipal pieces. A lower bound for $\mathfrak{P}_X$ alone is not a stop certificate unless one also knows $B_X \geq X^{-o(1)}$. The stated order tests the cheapest kill conditions first. □

*Remark* 4.2 (What counts as forward progress). A proof of Gate W is L1 bookkeeping progress. A proof of Gate P or Gate X for the actual growing census is closer to an arithmetic estimate, but it becomes L2 only after its exact role in the full growing fixed-$h_0$ coefficient and all remaining reassembly losses is verified.

## 5 Independent and fallback routes

### 5.1 Signed filter before the positive majorant

The positive census replaces

$$\mathfrak{m}_{K,X}(r)\, A_{\beta,X}\, \zeta_{\beta,r}\, S_{\beta,X}(r)$$

by its absolute envelope. If Gate P or Gate X is too large, the next route is to keep this coherent product and seek a weighted two-form estimate before summing over $r$. Such a theorem must retain:

- the signed reconstruction multiplier, not only its modulus;
- the $r$-dependent unit phase and affine character;
- the actual Möbius factors inside the resolved $z$-fiber; and
- coherent outer reassembly across frequency-free branches.

A theorem for a factorable surrogate with independent $r$ and $\beta$ is not automatically a theorem for this literal product.

### 5.2 Dominance and determinant energy

TPC-95 gives a sharp phase-minimization identity and a compressed shared-target census [4]. A successful continuation needs an actual determinant-energy lower bound with loss $\lambda_D < 1$, not merely an ambient or random-model estimate. It must then satisfy the independent compatibility

$$\lambda_D \leq 2\eta_Z.$$

This route can coexist with the resonance program; gains may be combined only when they apply to disjoint costs rather than the same cancellation twice.

### 5.3 Tauberian translation

TPC-96 proves the exact anchored identity

$$S_N = (\rho + N)L_{\rho,N} - \sum_{m<N} L_{\rho,m}$$

and isolates the translation defect. Under the recorded base condition, terminal averaging is equivalent to the correct scale

$$\frac{\rho}{N} L_{\rho,N} = o(1),$$

not to an unqualified $L_{\rho,N} = O(1)$ [5]. Any future logarithmic primitive estimate must be inserted with its actual anchor $\rho$.

### 5.4 Projection and compression

TPC-97 proves that a constant-preserving orthogonal projection keeps the zero mode fixed while lowering energy [6]. Therefore it gives a conservative upper-bound interface, not a source of new normalized cancellation. If $P\mathbf{1} \neq \mathbf{1}$, the exact constant defect must be budgeted at the critical $J_X^{-1}$ scale.

### 5.5 Full-block reassembly

TPC-18's full-block return and primitive determinant dispersion remain independent [1]. Closing every resonant low-window branch would still leave:

1. the long generic affine sector;
2. the high-frequency reconstruction tail at the chosen $K, R_X$;
3. block-to-physical normalization;
4. determinant-energy compatibility; and
5. the final strict endpoint loss.

*Remark* 5.1 (Fallback is not failure). If the positive route hits a sharp counterexample, the result is a valid negative theorem about that majorant. It directs the program to signed arithmetic or full-block reassembly; it does not imply that the original fixed-shift problem is false.

## 6 Scale and exponent ledger

The current physical scales are

$$Q_X = X^{267/400+o(1)}, \qquad J_X = X^{133/400+o(1)}, \qquad q_X \asymp Q_X.$$

The endpoint-scale branch return is

$$Q_X^2 = X^{534/400+o(1)}.$$

### 6.1 What TPC-101 saves on a long fiber

The previous worst-eigenvalue factor was

$$q_X^{1/2} = X^{267/800+o(1)}$$

apart from logarithms. On a resolved branch of length $N \geq J_X$, TPC-101 gives

$$g_{q_X}(H_{q_X}(N)) \leq \sqrt{\frac{2q_X}{N}} \leq X^{67/400+o(1)}.$$

The difference is

$$\frac{267}{800} - \frac{67}{400} = \frac{133}{800}.$$

Thus the spectral return gains a factor

$$J_X^{1/2} = X^{133/800+o(1)}$$

on that length block.

*Remark* 6.1 (Support qualification). This gain is rigorous *on branches known to satisfy* $N \geq J_X$. The full literal packet also contains short fibers. No global $J_X^{1/2}$ saving may be entered until their weighted mass is separately returned.

### 6.2 No new physical loss

The function $g_q(H)$ is part of an exact finite-group norm bound. It is not a new smoothing parameter and spends no physical exponent by itself. The obligations

$$\mathfrak{P}_X, \qquad W_X, \qquad \mathfrak{X}_X$$

must be estimated in the already normalized physical frame. Artificially enlarging the support to make a large-sieve theorem applicable would create a new loss, which must be entered explicitly in $\Lambda_{\mathrm{phys}}$.

### 6.3 Endpoint stop

The final rule remains

$$\Lambda_{\mathrm{phys}} < \frac{1}{400}.$$

A derivation reaching equality does not close the endpoint. Subpower factors, logarithms, and fixed comparison constants are absorbed in $X^{o(1)}$; a fixed positive power is not.

| Quantity | Present target | Accounting rule |
|---|---|---|
| $\mathfrak{P}_X$ | $X^{o(1)}Q_X^2$ | Positive principal mass; centered cancellation cannot pay it |
| $W_X$ | $X^{o(1)}$ | Pointwise normalized atom cap; any power loss is physical |
| $\mathfrak{X}_X$ | $X^{o(1)}Q_X^2$ | Already includes the width factor $g_q(H)$ |
| $\lambda_D$ | $\leq 2\eta_Z$ | Independent determinant/zero-mode compatibility |
| $\Lambda_{\mathrm{phys}}$ | $< 1/400$ | Strict endpoint; equality is a stop |

Table 2: Nonduplicated exponent ledger.

| Gate | Go certificate | Stop certificate | Valid response |
|---|---|---|---|
| Atom cap | $W_X \leq X^{o(1)}$ from literal factors | $W_X \geq X^{\delta}$ on active mass, with no offset | Atom-cap transfer stops; reopen the exact diagonal ledger or restore signs |
| Principal mass | $\mathfrak{P}_X \leq X^{o(1)} Q_X^2$ | $B_X \mathfrak{P}_X \geq X^{\delta} Q_X^2$ for the positive census | Three-ledger bound stops; sharper filter or signed route remains |
| Cross cells | $\mathfrak{X}_X \leq X^{o(1)} Q_X^2$ | $B_X \mathfrak{X}_X \geq X^{\delta} Q_X^2$ in the current bound | Three-ledger bound stops; aggregate exact maps or restore signed arithmetic |
| Generic sector | Growing fixed-$h_0$ two-form estimate | Only averaged, frozen, support-only, or random-model gain | Keep result at L0/L1 |
| Endpoint | $\Lambda_{\text{phys}} < 1/400$ | $\Lambda_{\text{phys}} \geq 1/400$ | Do not claim closure |

Table 3: Route decisions are statements about a method, not about the truth of the twin-prime conjecture.

# 7 Stop–go criteria

## 7.1 Six-item claim audit

Before any new result is promoted, check:

1. Are the literal physical coefficients present?
2. Is one fixed nonzero $h_0$ present throughout?
3. Is the atomic quantity normalized in the physical frame?
4. Is the representation canonical or at least losslessly invertible?
5. Is the estimate on the actual active support?
6. Does the final physical ledger remain strictly below 1/400?

Failure of any item prevents a claim of final proof progress.

## 7.2 What a negative theorem can publish

A sharp family with $B_X \mathfrak{P}_X$ or $B_X \mathfrak{X}_X$ polynomially above the target would prove that the present three-ledger, principal-plus-absolute-remainder implementation cannot close. It would not prove the positive carrier itself large or exclude a sharper filter-sensitive estimate. A polynomial atom-cap failure would identify the exact coefficient responsible and stop only the atom-cap transfer. A projection counterexample would show why a compressed model loses the physical zero mode. Each is a valid mathematical conclusion when stated at its actual level.

*Remark* 7.1 (No inference to RH or prime pairs). Spectral terminology here concerns finite incidence operators and literal dynamical bookkeeping. It does not identify Riemann zeros with eigenvalues, prove the Riemann hypothesis, or imply a prime-pair asymptotic.

# 8 Recommended next batch

The next papers should test the cheapest literal gates before attempting a new deep cancellation theorem.

### TPC-103: normalized opened-atom cap

Write $w_C(j)$ from the TPC-93 source atom through TPC-94 content inversion and the TPC-100 $(\beta, z)$ refinement. Prove or refute

$$W_X \le X^{o(1)}.$$

Required outputs:

- an exact factor list with one global normalization;
- pointwise bounds for row, logarithmic, divisor, projector, and smooth factors;
- a finite inverse-provenance regression; and
- a polynomial-loss counterexample if the cap fails.

### TPC-104: principal mass by resolved length

Open

$$\mathfrak{P}_X = \sum_{\substack{\beta \in \mathfrak{B}_X^{\mathrm{nc}} \\ 1 \le N_\beta \le q_X}} \frac{2H_{q_X}(N_\beta)}{q_X - 1} \sum_{\substack{z \in I_\beta^* \\ q_X \nmid u_{\beta,z}}} w_{\beta,z}$$

before grouping equal multipliers. Use

$$\frac{H_q(N)}{q} \le \frac{1}{N}, \qquad N \ge 3,$$

together with the harmless exact conversion from $q$ to $q-1$, and return the remaining invertible $N = 1, 2$ mass literally. No ambient branch count may replace the displayed opened-atom mass.

### TPC-105: canonical affine-map aggregation

For invertible opened atoms, store

$$\omega_C(j) = s_C j + t_C, \qquad s_C = A_C m_C, \quad t_C = A_C h_0.$$

Aggregate cells with identical $(s_C, t_C)$ only after attaching a recoverable provenance bag. Split $\mathfrak{X}_H^\circ$ into:

$$\text{identical-map concentration} \quad + \quad \text{genuinely distinct-map incidence.}$$

The first is a new mass ledger; the second is the honest geometric door.

### TPC-106: width-weighted cross-map estimate

Attempt

$$\sum_H g_{q_X}(H) \sqrt{|\mathfrak{X}_{H,\mathrm{distinct}}^\circ|} \ll X^{o(1)} Q_X^2$$

using an additive Fourier dual, a restricted affine large sieve, or a determinant-aware incidence theorem. The theorem must retain the actual map support and physical weights.

### Fallback sequence

If TPC-103 exhibits an uncompensated polynomial atom loss, or if TPC-104 proves $B_X \mathfrak{P}_X \ge X^\delta Q_X^2$, do not spend another paper on positive cross-cell dispersion. Move directly to:

1. signed filter-before-majorant reconstruction;
2. divisor-twisted character or additive Fourier analysis;
3. comparison with the TPC-95 dominance-energy route; and

4. a full TPC-18 reassembly audit.

**Proposition 8.1** (Decision order)**.** *The order TPC-103 → TPC-104 → TPC-105 → TPC-106 minimizes the risk of proving an expensive theorem that cannot close* (8)*.*

*Proof.* TPC-103 tests whether the diagonal transfer is available. TPC-104 tests a term that no centered theorem can cancel. Only after both pass is the cross-cell term the unique positive resonance obstruction. TPC-105 then removes artificial identical-map duplication before TPC-106 attacks genuine dispersion. □

# 9 Finite audit certificate

The script `experiments/tpc102_audit.py` checks:

1. the disjoint branch partition
$$q_X \mid \Omega, \qquad q_X \nmid \Omega,\ q_X \mid u, \qquad q_X \nmid \Omega,\ q_X \nmid u;$$
2. deterministic opened-cell ledgers against the principal-to-diagonal radical in (8);
3. the exact endpoint $g_q(H_q(1)) = g_q(H_q(2)) = 0$;
4. the length inequality $g_q(H_q(N)) \le \sqrt{2q/N}$ for $N \ge 3$; and
5. the rational exponent identities
$$\frac{267}{800} - \frac{67}{400} = \frac{133}{800}.$$

This certificate tests the logic and exponent bookkeeping of the audit. It does not test (9), (10), or (11) on the growing literal packet.

# 10 Claim boundary

## Established in this synthesis

- The dependency table separates exact algebra, literal branch returns, conditional interfaces, and missing arithmetic inputs.
- The conductor-one and nonconstant $q_X \mid u$ branches are disjoint and already returned at $X^{o(1)}Q_X^2$.
- The remaining invertible positive resonance carrier obeys the three-ledger master inequality (8).
- Principal mass, a subpower opened-atom cap, and width-weighted cross-cell excess are sufficient for complete positive resonance closure.
- On a branch with $N \ge J_X$, the exact fixed-filter theorem saves $J_X^{1/2}$ over the former worst-eigenvalue factor.
- The recommended order audits cheap literal kill conditions before attempting a deep cross-cell theorem.

These are L1 synthesis and route-selection results resting on the cited literal papers.

### Not established

- None of $\mathfrak{P}_X \le X^{o(1)} Q_X^2$, $W_X \le X^{o(1)}$, or $\mathfrak{X}_X \le X^{o(1)} Q_X^2$ is proved here for the growing packet.
- Positive resonance closure is not the same as a signed low-window estimate.
- The long generic affine sector and the high-frequency tail are not closed by (8).
- TPC-18 full-block reassembly, primitive determinant dispersion, the actual determinant lower bound, and strict endpoint loss remain independent.
- No growing fixed-$h_0$ affine Möbius cancellation theorem, sieve-parity breakthrough, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime conclusion is proved or implied.
- No relation between Riemann zeros and a dynamical spectrum is claimed.
- Nothing is specialized to $h_0 = 2$.

### Current honest verdict

The route is still open and has narrowed substantially. It has not reached a pure negative result, because two literal branches are closed and the crude worst-eigenvalue bound has been replaced by the exact width-sensitive norm. It has also not reached L2, because the three master ledgers and the generic signed sector remain unproved. The next decisive question is local and falsifiable: whether the actual normalized opened atoms have subpower size.

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